

AquaGuard – One-Page Technical Progress Report

Team: 404 Found

Technical Progress

The AquaGuard team has completed the initial technical planning and system design for the proposed IoT-based water-quality monitoring system.

- ESP32 has been selected as the main IoT controller.
- The required pH, TDS/EC, turbidity, and temperature sensors have been identified for water-quality monitoring.
- A GPS module has been selected to record the location of each water-quality measurement.
- The overall data flow has been designed as Sensors → ESP32 → GPS + Measurement Data → Cloud → Mobile Application.
- Flutter has been selected for the mobile application and Firebase for cloud data storage.
- Python has been selected for data analysis, with Google Maps/OpenStreetMap planned for location mapping.
- The mobile application structure has been planned to display water-quality measurements, locations, historical data, graphs, and alerts.
- An offline data storage and synchronization mechanism has also been planned for areas with limited internet connectivity.

➤ Current Status

System architecture and technology selection completed. The team is currently moving towards hardware integration, sensor testing, cloud implementation, and mobile application development.

➤ Next Technical Steps

ESP32 & Sensor Integration → GPS Integration → Firebase Database → Flutter Application → Testing & Integration

➤ GitHub Repository

GitHub Link: <https://github.com/ANafeesh/AquaGuard-IoT>